

Felipe Pombal Fritsch

London, UK || +55 (21) 97116-0567 || felipe.fritsch@columbia.edu

EDUCATION

University of Oxford, Mathematical Institute (Graduate)

Oxford, UK

- MSc Mathematical and Computational Finance (2025–26)
 - Dissertation: “Deep Learning Models of Loan Portfolios and Asset-Backed Securities”
Advisor: Prof. Justin Sirignano | [GitHub](#)

Columbia University, Columbia College (Undergraduate)

New York, NY

- B.A. in Applied Mathematics and B.A. (Hons) in Economics — Graduated May 2020
 - GPA 3.8/4.0 (Major 3.9/4.0); Economics Departmental Honors
 - Honors Thesis: “Bounds on the Dynamic Causal Effects of Monetary Policy: An Analysis using the Federal Reserve’s Textual Repository” — awarded the maximum grade

Instituto de Matemática Pura e Aplicada (IMPA)

Rio de Janeiro, Brazil

- Advanced studies in Machine Learning, focusing on deep learning and the foundations of LLMs / RL (early 2024)

The British School of Rio de Janeiro

Rio de Janeiro, Brazil

- International Baccalaureate: 44/45 | Gold Medal, Brazilian Astrophysics Olympiad (2015)

EXPERIENCE

Gávea Investimentos

Rio de Janeiro, Brazil

Quantitative Researcher — Sep 2024 – Aug 2025

- Built forecasts, nowcasts and fair-value signals across macro variables and asset classes in 30+ countries, working directly with PMs in a discretionary-macro investment process.
- Led development of a multi-agent RAG platform adopted firm-wide by PMs and economists, with a key role in writing the model-related code and developing the LangChain pipeline.
- Combined contextual embeddings in Pinecone with hybrid retrieval and reranking, using LangChain/LangGraph for query decomposition and document selection alongside Text2SQL integration.
- Refined the platform through four prototypes using investment users’ questions, with DeepEval evaluations in CI and LangSmith for observability; maintained rate limits, separate dev/prod environments and data refresh jobs.

Atmos Capital

Rio de Janeiro, Brazil

Equity Analyst — Feb 2023 – Jan 2024

- Built quantitative tools and internal web applications for data-analysis and trading teams using Python backends, JavaScript/HTML and REST APIs; piloted a systematic signal effort.
- Owned fundamental company coverage and full financial analysis; presented and defended recommendations on new investments and position reductions.

AQR Capital Management

Greenwich, CT

Quantitative Research Analyst — Jul 2020 – Oct 2022

- Developed new signals and improved existing ones for systematic credit strategies, spanning value, momentum, defensive and reversal styles across single-name corporate bonds and CDS.
- Focused extensively on developing structural and empirical models of default prediction and rating downgrades, evaluating alternative vendor models against internal approaches.
- Derived a robust estimate of corporate debt-to-equity elasticity to translate potentially additive/orthogonal equity factors into our credit models.

Quantitative Research Summer Analyst — Jun 2019 – Aug 2019

- Investigated corporate-bond returns around earnings announcements to assess whether the well-documented effects in equities carried over to credit markets.

LEADERSHIP & ACTIVITIES

Columbia Quant Group

New York, NY

Vice President — Aug 2019 – May 2020

- Organized recruiting and educational events introducing STEM students to quantitative finance.

Columbia University, Mathematics Department

New York, NY

Teaching Assistant — Aug 2017 – Dec 2018

- Taught and graded Calculus I–III, Linear Algebra and ODEs; ran a twice-weekly help room.

SKILLS & PERSONAL

Languages: English & Portuguese (fluent), Spanish (intermediate) | Programming: Python (NumPy, pandas, PyTorch), C++, R, SQL, Git; JavaScript/HTML

AI: LangChain/LangGraph, RAG, Text2SQL, Pinecone, contextual embeddings, DeepEval, LangSmith